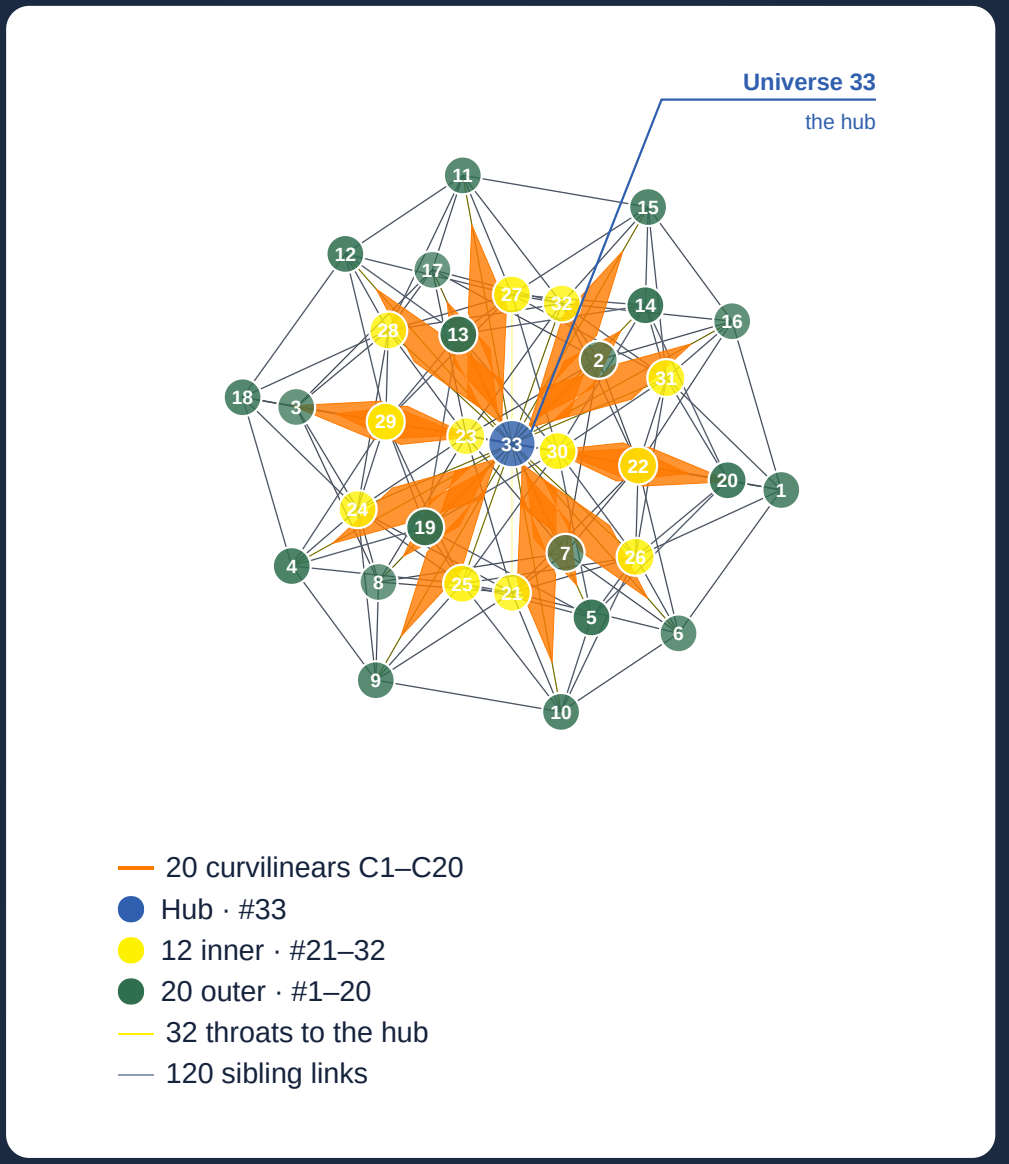
AI33-MPOPT · QUANTUM-GRAVITY PROGRAM · 2026

One Operator. Real Numbers. Scored Against Data.

Testing whether a 33-universe icosahedral geometry can bridge quantum mechanics and gravity — scored result by result.

Rolando Rivero · Independent Researcher · AI33-MPOPT Framework



CONTEXT

A one-hundred-year standoff

General relativity describes gravity as smooth geometry. Quantum mechanics describes matter as discrete states. Both are spectacularly confirmed — and they refuse to merge. Every mainstream road starts from a singular Big Bang and tries to quantize gravity on top of it.

AI33-MPOPT inverts the starting point: begin from a finite geometric structure — 33 universes on an icosahedron — and ask whether gravity-like behavior falls out of a quantum spectrum.

This program tests one bridge, scored result by result.

1915

General Relativity

gravity as geometry

1925

Quantum Mechanics

matter as spectra

~100 yrs

No accepted merger

the deepest open problem in physics

METHOD

Every claim wears its status

GOLD — PROPOSAL

Framing and vision under test. Stated plainly as unproven. Example: the corridor, the pre-inception smoothness idea.

GREEN — COMPUTED

Verified, reproducible numbers with statistical controls. Example: the 11-mode band, the multiplet structure.

RED — FITTED

External data. $p = 0.781 \pm 0.013$ from a 96,000-sample MCMC fit against JWST chronometer ages.

"A formula that appears all at once, fully formed, is exactly the pattern this investigation learned to distrust."

COMPUTED

The super formula is an operator

$T_p = L + pK + (1 - p)Q$

An explicit 33 × 33 matrix — not a metaphor, not a placeholder. Verify each piece and you have verified the whole. **L is the canonical symbol for the Laplacian term throughout this deck** (earlier versions wrote D — same matrix).

T is positive definite at p = 0.781 (λ_{min} = 0.978), so $G = T^{-1}$ exists — verified, not inferred from a plot.

K here is in adjacency form; with K in Laplacian form the same operator gives λmin = 0.063 — the convention must be stated with the value. 12 apexes = the icosahedron's vertices (pull, dark matter). 20 curvilinears = its faces (push, dark energy).

L · VERIFIED

Connectivity. Graph Laplacian of the 33-universe network. Pure geometry — zero modeling choices.

K · 1 FLAG

Throat coupling. Hub coupling 1.0 vs sibling 0.1(1−p)/32 — archive rule. Overlap g(i,j): flagged choice.

Q · ROBUST

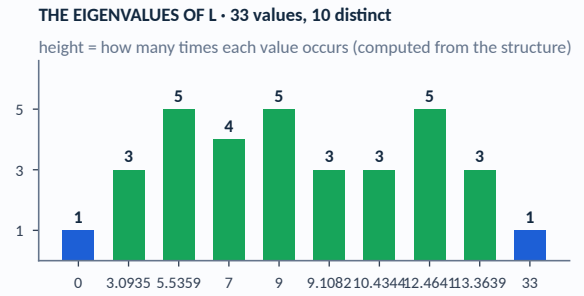
Capacity. Degree-normalized throughput. Three forms tested — the band survived all three.

p · FITTED

The seam. The one external number: 0.781 ± 0.013 from JWST cosmology. Its dual role is an assumption.

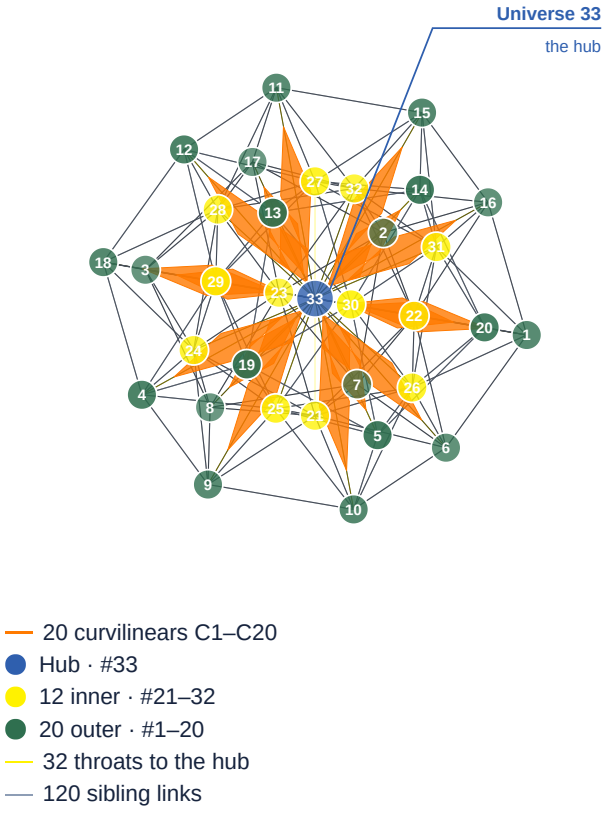
THE OPERATOR TP = L + PK + (1 − P)Q, BUILT FROM ITS THREE PIECES





COMPUTED

The geometry is exact, not assumed



1 + 12 + 20

one hub (our universe), twelve dark-matter apexes, twenty dark-energy curvilinears of a Platonic icosahedron

152 edges

30 vertex–vertex + 60 vertex–face + 30 face–face + 32 throats — every count fixed by the solid itself

0 free parameters in L

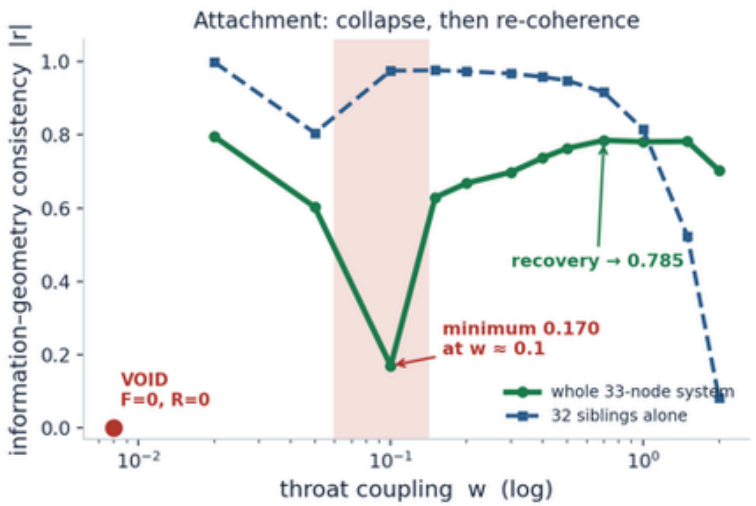
the Laplacian is standard mathematics; nothing about connectivity is invented

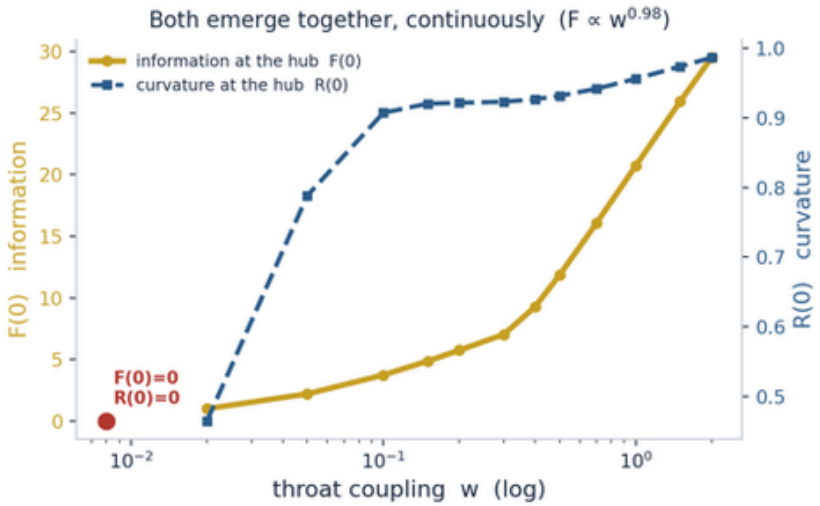
12 apexes circular, direct line to the hub — dark matter — U21–32
32 siblings the two together — the 32-multiverse, complete on its own
32 throats centre to every sibling — "throat coupling" is their strength

20 curvilinears three-sided regions between the apexes — dark energy — U1–20
1 hub Universe 33, ours — at the geometric centre, not on the surface
120 sibling links across the surface — 30 v–v + 60 v–f + 30 f–f

COMPUTED

The Void — lawfulness without relations





WHAT EXISTS IN THE VOID

Set the throat coupling to zero and the operator T becomes SINGULAR — it has no inverse, so there is no Green function and no propagator.

Every quantity vanishes exactly:
$F(0) = 0 \quad R(0) = 0$
$G_{00} = 0 \quad \max|G_{0j}| = 0$

No information. No curvature. No influence arriving. Not assumed — computed.

NO PROPAGATOR → NO 'HOW LONG'

Distance, travel time and causal order are all defined through the propagator. With no relations there is no propagator, so questions of duration have no answer in the void — not unknown, undefined.

THE SIBLINGS ARE ALREADY LAWFUL

While the hub sits at zero, the 32-sibling structure alone holds the law at 0.988. Lawfulness does not begin at inception; the hub enters a structure that is already coherent.

ATTACHMENT: COLLAPSE, THEN RE-COHERENCE

Weak coupling is disruptive: whole-system consistency falls to 0.170 at w ≈ 0.1. Above w ≈ 0.15 the structure re-coheres, rising to 0.785 by w ≈ 0.7. Inception has a threshold.

QUANTUM AND GRAVITY EMERGE TOGETHER

F(0) and R(0) leave zero at the same moment and rise together — F continuously, as w^0.98. Neither precedes the other; both appear as relations appear.

Interpretation, stated as such: the void is a lawful position with zero relations — where an information reading and a geometry reading are not yet distinguishable, because neither has anything to read.

FITTED

One number stitches gravity to quantum

$$p = 0.781 \pm 0.013$$

Paused-Gravity fit $H(z) = p \cdot H\Lambda CDM(z)$
emcee · 64 walkers · 96,000 samples · seed 42
against JWST chronometer ages

WHERE p STANDS

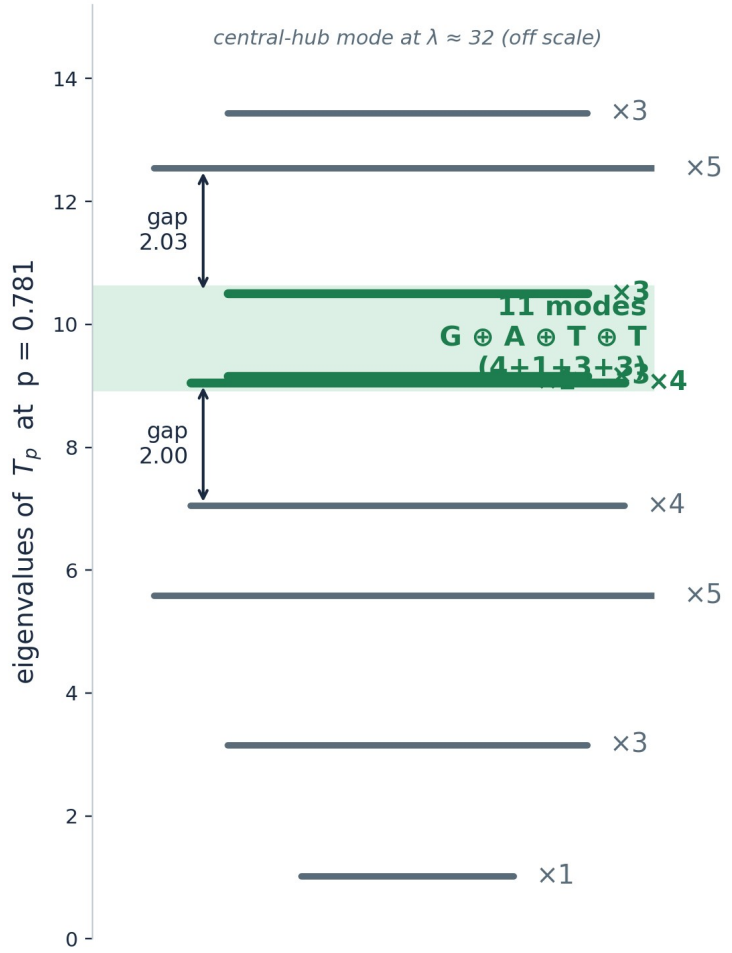
p is a fitted cosmological number. Reusing it inside a quantum operator is the program's working assumption: the seam where the two theories are stitched together, and the point future work will justify or replace.

A fixed-point self-consistency test lands at $p^* = 0.7577$ — within 1.8σ of the fitted value.

Superseded claims: the "derivation" $p = \exp(-F \cdot N)$ is circular (F itself is fitted) · the corridor bounds 0.5637 / 0.8009 trace to weakly-constrained runs and are carried as a proposal.

COMPUTED

Eleven modes, exactly, in a protected band



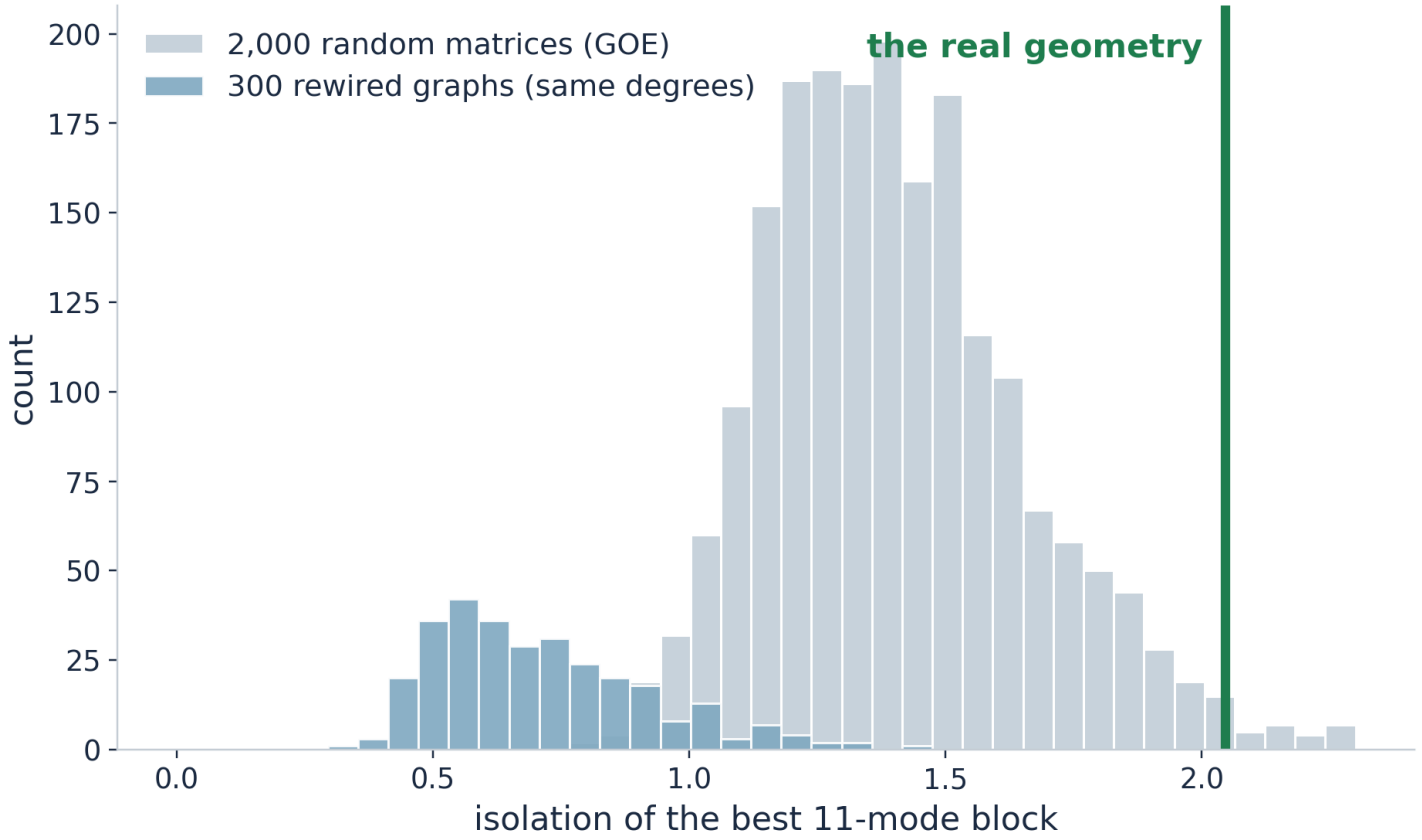
Diagonalizing T at the fitted p yields machine-precision icosahedral multiplets — and an isolated 11-mode band with gaps of ≈ 2.0 on both sides (mean spacing ≈ 0.98).

Its decomposition G ⊕ A ⊕ T ⊕ T (4+1+3+3) matches the icosahedral symmetry group exactly — the same multiplet content the archive reported, now reproduced in a fully independent rebuild with its own code and its own modeling choices.

Why 11 matters: an 11-dimensional invariant sector is the structural echo this program looks for between the 33-universe geometry and 11-dimensional M-theory. Echo — not equivalence.

COMPUTED

The band is not luck



1.7%

of 2,000 random spectra

reach the observed isolation

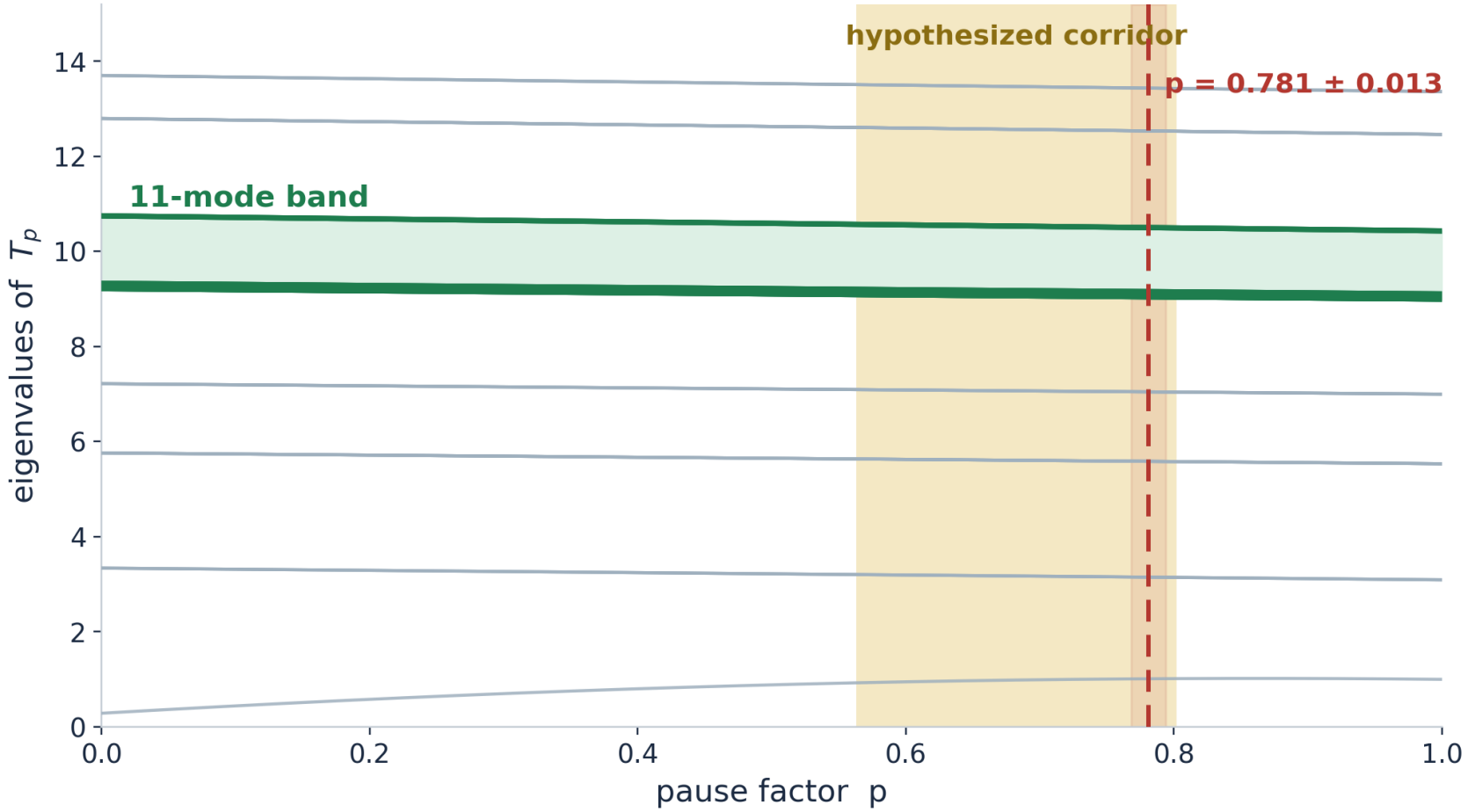
0 / 300

rewired graphs match it

same degrees, scrambled wiring

COMPUTED

The band survives the entire sweep



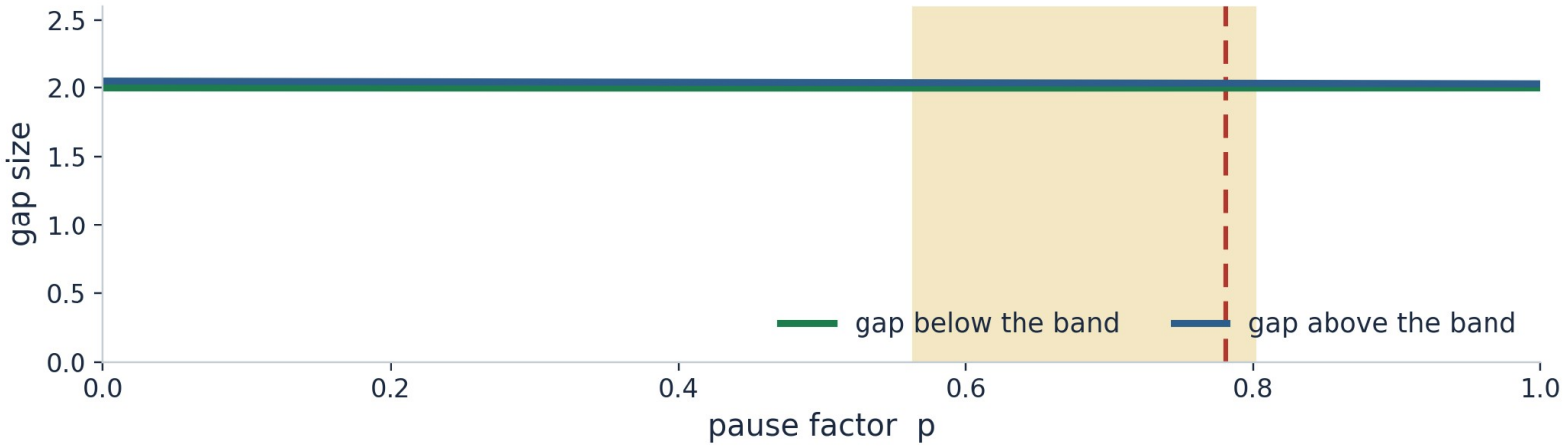
Every eigenvalue branch tracked across all p ∈ [0, 1]. No level crossing anywhere in the physical range. Band isolation drifts only 1.90 → 2.06 end to end.

Central-hub mode (λ ≈ 32) sits far above — throat-locked and p-inert.

PROPOSAL

The band belongs to the geometry

The band is a property of the icosahedral geometry itself: it survives the entire sweep of p from 0 to 1 with no level crossing, isolation drifting only 1.90 to 2.06 end to end. The spectrum shows no feature at the proposed corridor [0.5637, 0.8009], which is carried as a proposal.

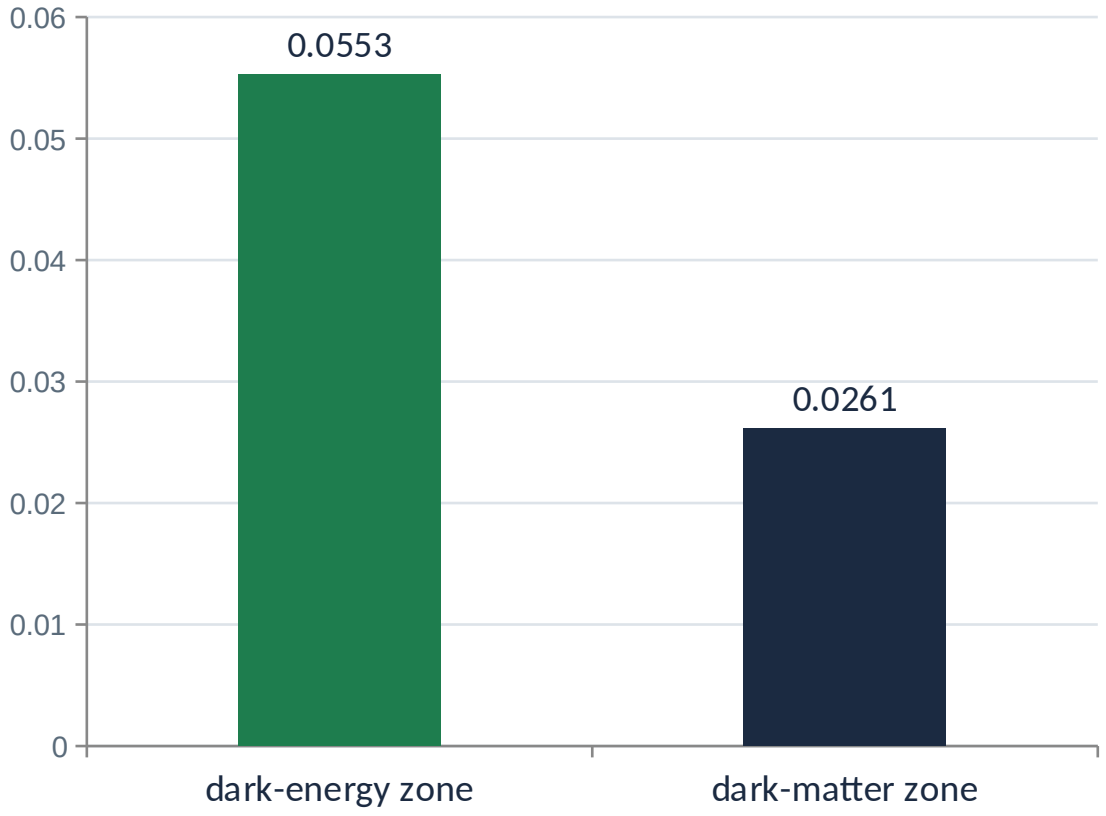


Consequence: if p does physical work inside T_p, it must act through the eigenvectors — exactly where the two curvature results on the next slides live.

RESULT 2 — EIGENVECTORS MEET CURVATURE

COMPUTED

The quantum side predicts which zone is more curved



Mutual information — a standard quantum-information quantity, computed from T_p's own eigenvectors, no invented formula — correctly picks the dark-energy zone as the more curved region.

2.12

ratio predicted

from eigenvectors

2.99

ratio measured

independent curvature

1.46

ratio, intrinsic

Ollivier on Tp weights

Tested first: raw density and density-weighted flux were both tested and eliminated. The three ratios above are not the same quantity: 2.12 is the mutual-information ratio plotted at left, while 2.99 and 1.46 are curvature ratios. Direction agrees throughout; magnitude is the open fit, and the archive recipe must be pinned.

COMPUTED

Group theory said five. Five appeared.

5 = 5

predicted = built

invariant 3-forms in the 11-mode sector — the object type carrying M-theory's flux

4 / 5

point the right way

as physical densities, four of five pick the correct more-curved zone (~50% for chance)

0 / 5

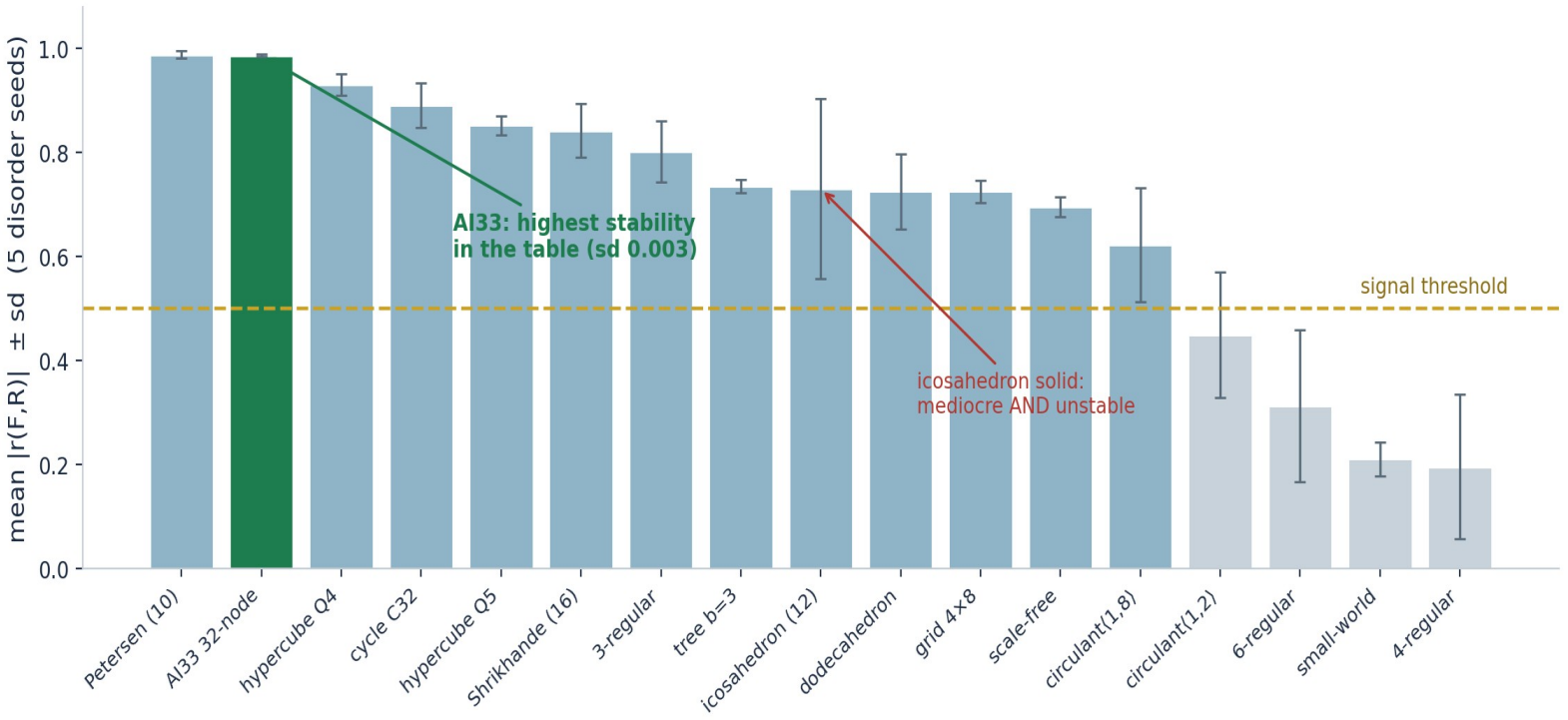
match the magnitude

direction is real progress; the size of curvature is still unresolved

These are not invented formulas — they are the literal tensors the framework's own symmetry predicted, built explicitly and tested against independently computed curvature for the first time. The missing magnitude is Wall #1 of this program.

COMPUTED

It holds on 13 of 17 graph families



Every value is the mean of five independent disorder seeds with its standard deviation. Single-seed values proved unreliable and were discarded.

With the hub at zero, the 32 universes already agree at 0.987 ± 0.003, and the law holds again once the hub attaches (The 33-Multiverse Map, page 57).

NOT A CURVE-FIT

13 of 17 families carry signal (mean |r| > 0.5, criterion set before testing). The coupling is a broad phenomenon, not a property of one graph.

AI33: 0.987 ± 0.003

Top of every family tested, and the most stable by a factor of two — the tightest spread in the table. Strong coupling is not unique to it: Petersen matches at 0.988 ± 0.007, on only 10 nodes.

TWO DIFFERENT OBJECTS

The 12-vertex icosahedron solid scores 0.730 ± 0.173. The 32-node construction is a distinct object.

THE STRUCTURE HOLDS — THE RECIPIENT DOES NOT

With the 33rd node attached at w ≈ 0.5, the sibling structure alone holds the law at 0.951 while the full system holds 0.761 — consistency degrades at the recipient, not in the established structure. Reverses at strong attachment (w = 2).

COMPUTED

Fifteen candidates eliminated

EIGHT GRAPH PROPERTIES

clustering · spectral degeneracy · distance-regularity · vertex-transitivity · diameter · girth · degree variance · algebraic connectivity

Best r = 0.42, p = 0.24. None survives correction.

SIX OPERATOR PROPERTIES

eigenvector localization · spectral gap of T · condition number · Green-function concentration · transport entropy · spectrum spread

Best r = 0.37, p = 0.16. None survives correction.

QUASIPERIODIC ORDER

Identical edge-weight multiset arranged three ways on a chain:
Fibonacci 0.225 · periodic 0.065 · random 0.472

Quasiperiodic order does not enhance the law — random beats it.

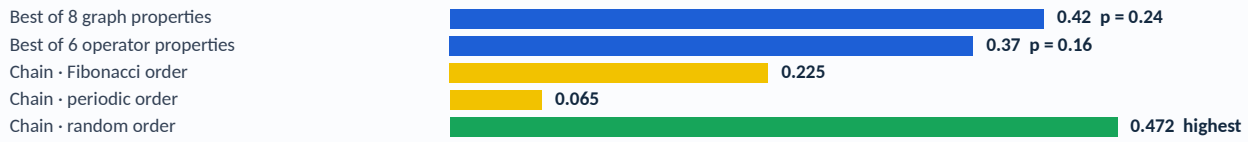
WHAT THIS SHOWS

A reproducible effect that resists fifteen standard descriptors is not a restatement of any of them. Slide 18 runs the different test, substituting a plain Laplacian within one graph, and the coupling survives unchanged: it sits at graph level.

WITH 106 GRAPH FAMILIES

Run on 106 families with a weighted graph Laplacian on 3D positions: four single properties survive correction (spectral gap, algebraic connectivity, diameter, Green-function concentration), and a random forest on all fourteen properties explains $R^2 = 0.61$ of the coupling under leave-one-out, against 0.14 for the best single property (shuffled-label test p ≈ 0.016, 60 shuffles).

FIFTEEN STANDARD DESCRIPTORS, EACH COMPARED WITH THE LAW

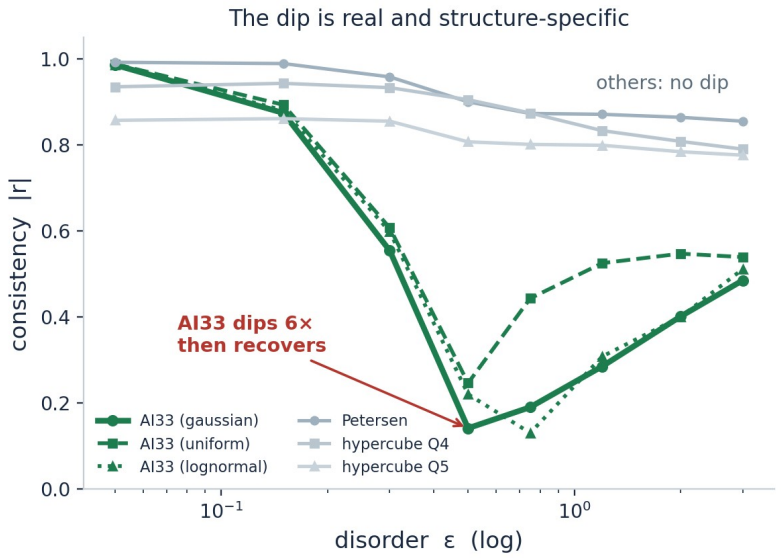


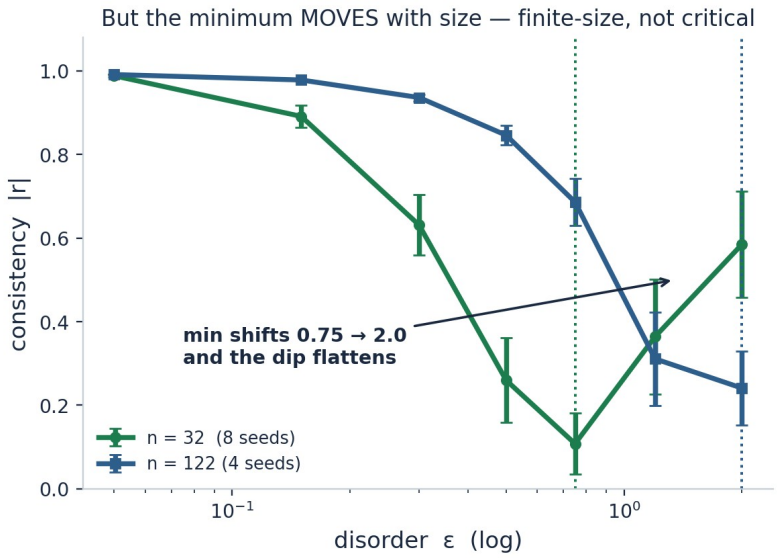
Eight graph and six operator properties: best r = 0.42 and 0.37, none survives correction.

One set of edge weights, three arrangements on a chain: random 0.472 · Fibonacci 0.225 · periodic 0.065.

COMPUTED

A consistency dip — real, but finite-size





WHAT THE SIZE TEST KILLED

A genuine phase transition sharpens toward a fixed critical value as the system grows. This does the opposite: subdividing the icosahedral structure from 32 to 122 nodes moves the minimum from ε ≈ 0.75 to ε ≈ 2.0 and makes the dip shallower. That is a finite-size crossover — larger systems average over more edges and resist per-edge disorder — not a critical point. The word ‘critical’ does not apply and is not used.

WHAT SURVIVES

The dip replicates under gaussian, uniform and lognormal noise, and is not a clipping artifact: it appears before any weight is clipped, and survives with strictly-positive weights needing no clipping at all.

THE FIRST REAL SEPARATION

Petersen and the hypercubes match AI33 on coupling strength and stability, and every invariant tested failed to tell them apart. Under disorder they separate: AI33 dips 6× and recovers; the others show no dip at all. This remains the only measured property that distinguishes them.

WHERE IT STANDS

A finite-size crossover, reproduced under three noise types; the only measured property that separates AI33 from Petersen and the hypercubes.

PROPOSAL

One living object, four shadows

The vision: the pieces were never separate. Spectrum, eigenvectors, information and curvature are not connected by extra equations — they are different representations of one dynamical geometric object, T_p. Even the observer is inside the geometry: node 0 is not outside the system looking in; it IS the hub the whole structure focuses on.

λ_n

SPECTRUM

energy levels — the quantum face

Ψ_n

EIGENVECTORS

structure — where p acts

$F(i) = \Sigma_j I(i,j)$

INFORMATION

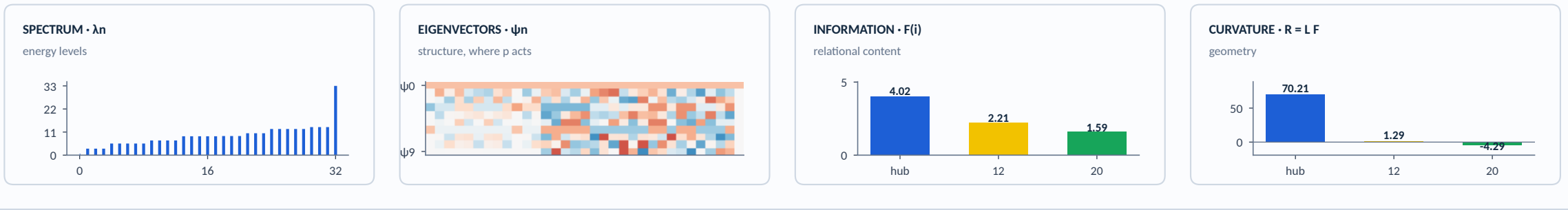
relational content — the candidate substance

R

CURVATURE

geometry — the gravity face

ONE OBJECT, FOUR SHADOWS — COMPUTED ON THE 33-DOMAIN STRUCTURE (L WITH SHIFT 0.5)



Historical pattern of every great unification: A → B becomes A = B seen from different viewpoints. Maxwell (electricity = magnetism), Einstein (gravity = geometry), quantum theory (wave = particle). The target here is a representation theorem — **measured on the next page, where two of the four representations turn out to be one.**

COMPUTED

Lawfulness, measured. Two shadows were one.

Slide 17 claims four representations of one operator. **Lawfulness** names their mutual agreement — six pairings between λ, Ψ, F and R. Five had never been computed. They have now.

λ ~ Ψ	λ ~ F	λ ~ R	Ψ ~ F	Ψ ~ R	F ~ R
+0.939	−1.0000	−0.586	−0.940	−0.397	+0.582

TWO REPRESENTATIONS ARE ONE

λ ~ F = −1.0000

at machine precision, every seed. The heat-kernel diagonal *is* the information field up to sign. “Four shadows” is three — and the count depends on a node-level choice the deck never stated.

THE CONTROL

+0.561 vs +0.582

R rebuilt from a **plain Laplacian** — no K, no Q, no p — matches T_p within seed noise. The coupling does not require the framework's operator. It is a graph property. Not a sixteenth invariant — the fifteen were predictors across families; this substitutes the operator within one graph, and is the first test that reproduces the coupling instead of failing to explain it.

WHAT SURVIVED

p = 0.0010

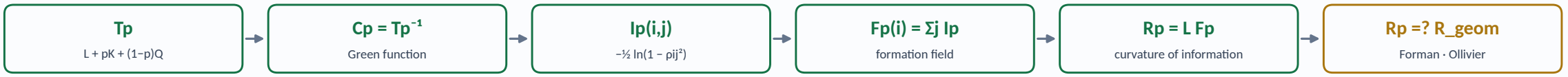
permutation test, 2,000 shuffles. The independent rebuild reaches +0.977 ± 0.007 at ε = 0.01 against the archive's 0.987 ± 0.003. The effect is real; its attribution has moved.

32-node structure · ε = 0.05 · 5 seeds · resistance metric · exact W_1 · **t = 1 × median transport time**. Values are **signed** correlations: the magnitude is the agreement, the sign is convention. r(F,R) runs +0.950 → +0.578 → −0.980 as t goes 0.25× → 1× → 4×, and reverses on an MI distance; elsewhere this deck quotes |r| = 0.987 at low disorder. **Full run: companion file AI33-MPOPT · Lawfulness.**

The complete super formula

THE OBJECT	$T_p = L + pK + (1-p)Q = \Sigma_n \lambda_n \lvert\psi_n\rangle\langle\psi_n\rvert$	computed · verified · controlled
INFORMATION REPRESENTATION	$C_p = T_p^{-1}$, $I_p(i,j) = -\tfrac{1}{2} \ln(1 - \rho_{ij}^2)$	Green-function recipe — documented, replaceable
FORMATION FIELD	$F_p(i) = \Sigma_j I_p(i,j)$	the eigenvector-borne field that feels p
GEOMETRIC REPRESENTATION	$R_p = L F_p$ · transport measures μ = rows of e^{-tT}, **t = 1 × median transport time**	curvature as the Laplacian of information — t must be declared (slide 18)
THE OPEN IDENTITY	$R_p \overset{\text{def}}{=}$ R_geom (Forman / Ollivier / …)	the theorem to prove or refute — this is the merger

THE CHAIN FROM THE OPERATOR TO THE OPEN IDENTITY



COMPUTED

The identity, tested — and what it exposed

F = diag(T_p) FAILS

the elegant spectral-diagonal candidate reduces to node degree — the density family already ruled out. It picks the wrong zone.

MI IS RECIPE-DEPENDENT

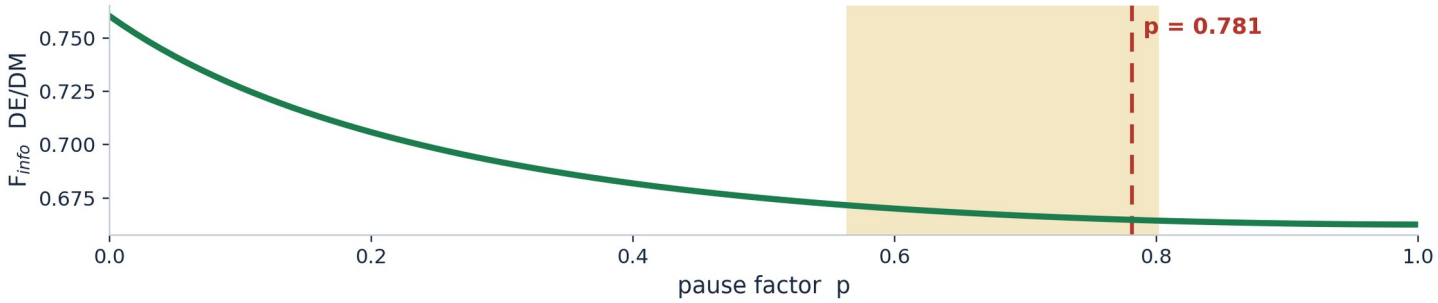
a documented Green-function MI gives near-parity (0.95), not the archive's 2.12 — the archived construction must be pinned.

CURVATURE SPLITS TOO

Forman: dark-energy more curved (1.41, direction agrees). Ollivier-Ricci: parity (1.00). The “2.99 measured” needs its definition named.

THE MODEL IS TOO PERFECT

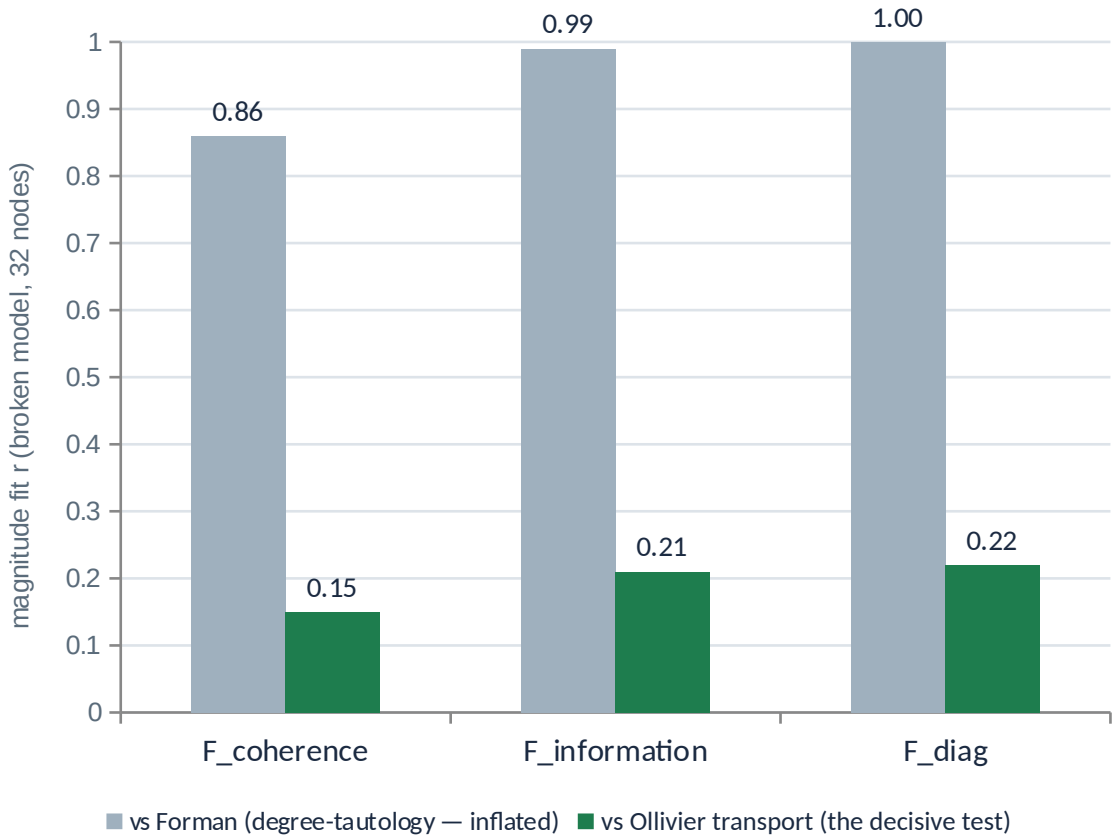
icosahedral symmetry collapses all 32 siblings into TWO equivalence classes — a magnitude fit has only two data points. The answer is not missing; the symmetry is hiding it.



The one clean positive: the information field responds smoothly to p while the spectrum stays silent — the seam truly lives in the information representation.

COMPUTED

The symmetry broke. The target moved.



CLOSURE: RESULT

p = e^(−32(F)) returns 0.449 at p = 0.781 and iterates to zero — the operator does not self-generate its pause factor.

COHERENCE: DIRECTION RESULT

F_coh picks the dark-matter zone (0.93) — third documented construction to contradict the archive's DE claim.

MAGNITUDE UNLOCKED

ε-weighted edges → 33 distinct eigenvalues. The 2-point degeneracy is gone; 32 real data points now exist.

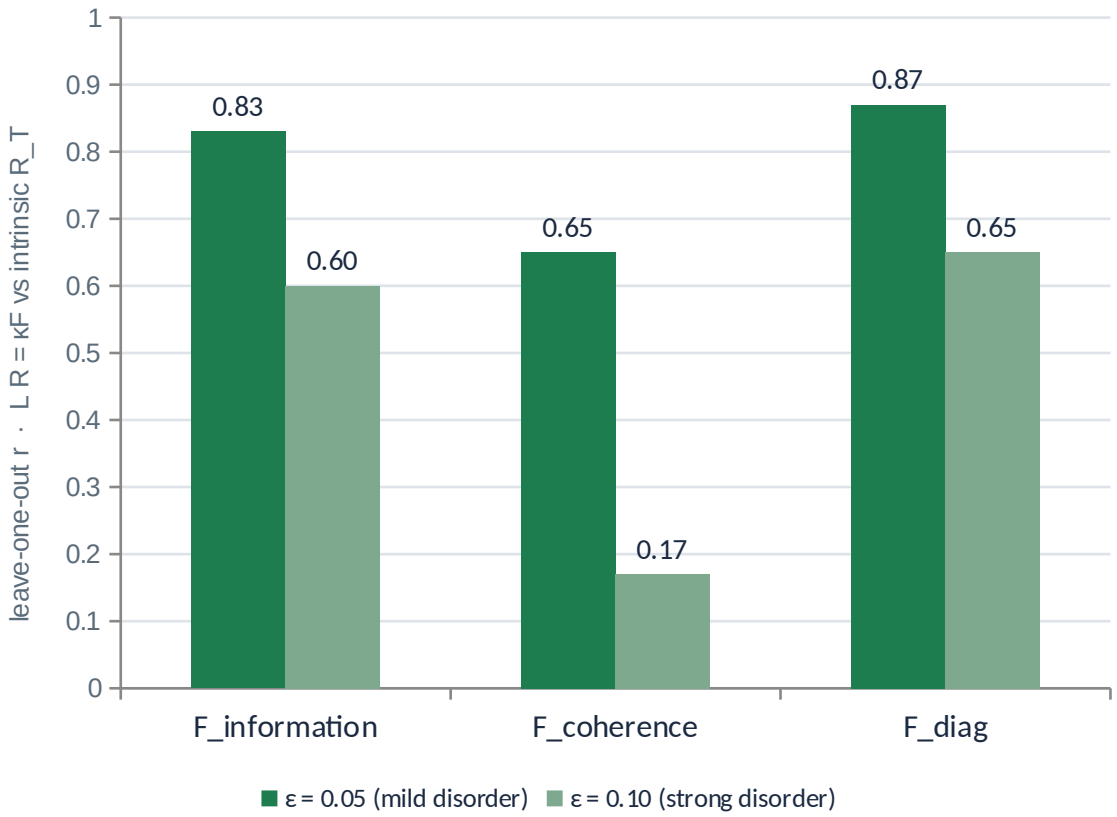
THE TARGET IS UNDEFINED

Forman and Ollivier curvature agree at only r ≈ 0.2 with each other. High Forman fits are degree-tautologies; vs transport curvature every candidate fails. The missing theorem is R, not F.

NEW COMPUTATION — THE THEOREM RUN

COMPUTED

The identity survived — approximately, with a twist



THE FRAMEWORK SELECTS ITS OWN R

intrinsic curvatures from T_P's transport agree at r = 0.88–0.996 across scales and constructions — the borrowed rulers agreed at only 0.2. R is defined — see slide 19.

THE IDENTITY HOLDS ON THE ORDERED GRAPH

L R = κF: best channel r = 0.84 (LOO 0.83), permutation p < 0.0005 (0 of 2,000 shuffles). Strength follows graph order.

κ — A SIGN TRANSITION

κ < 0 on the hub-centric icosahedron, κ > 0 on hub-free 3-regular graphs (5/5 seeds) — the repulsive sign is architecture-induced, not universal. The mechanism is mapped on the next slide: local clustering flips the sign.

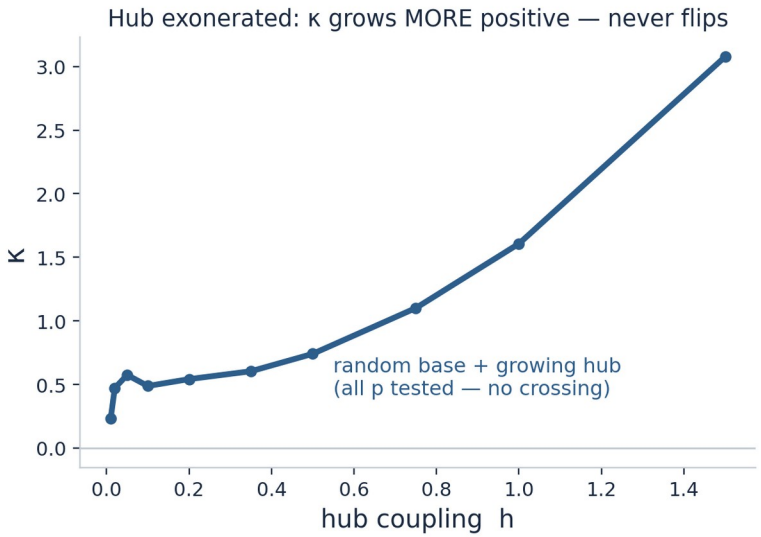
SOURCE IDENTIFIED: INFORMATION

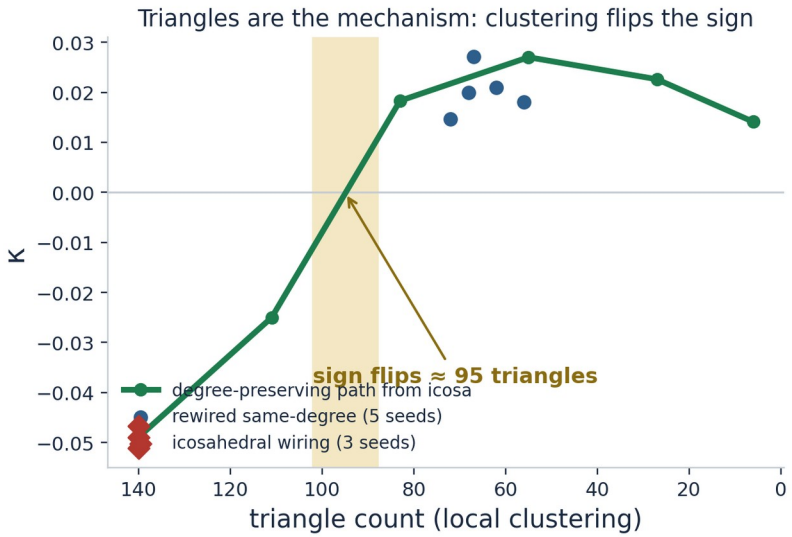
kill test on degree-constant graphs: F_diag eliminated by construction, yet the identity survives (5/5 seeds, mean LOO r = 0.64, perm p ≤ 0.022); partial correlations agree. Next: separate F_coh from F_mi.

NEW COMPUTATION — THE FLIP RUN

COMPUTED

Geometry selects the sign. Triangles are the switch.





NEAR-EXACT ON THE ICOSAHEDRON

With symmetry broken, the law runs at r = 0.96–0.98 on the hubless icosahedral geometry. The identity is most exact on precisely the geometry the framework chose.

HUB EXONERATED

Growing a hub on a random base amplifies κ upward at every strength and every p tested — no crossing anywhere. The hub amplifies whatever sign the geometry sets.

WIRING CONVICTED — 8/8

Icosahedral wiring: κ < 0 in 3/3 broken seeds. Identical degree sequence, rewired: κ > 0 in 5/5. Same degrees, opposite sign — the wiring itself decides.

THE CROSSING ≈ 95 TRIANGLES

Along a degree-preserving path from icosa (140 triangles) to random (≈65), κ rises monotonically and flips sign near 95. Local clustering is the mechanism — stated as a computed pattern, with the theorem still to prove.

NEW COMPUTATION — THE EXACTNESS RUN

COMPUTED

On the icosahedron, the law is exact at first order

$$L_0R_1 + L_1R_0 = \kappa_0 (F_1 - \langle F_1 \rangle) \qquad \kappa_0 = -0.0506$$

The linearized identity at the symmetric icosahedral point — the precise theorem target. The transport term L_1R_0 is essential: dropping it destroys the identity (an error made and corrected inside this program; recorded so successors avoid it).

r → 0.9996

as ε → 0

fit quality at ε = 0.005/0.01/0.02: 0.9996 / 0.9984 / 0.9936 — converging to exact

ONE κ, TWO ROADS

form-independence

R-side and F-side regressions converge to the same κ_0 = −0.0506 — the signature of an identity, not a fitting choice

5.6·ε

residual scaling

residual/signal = 0.113 / 0.280 / 0.533 at ε = 0.02 / 0.05 / 0.10 — all deviation is second-order disorder

+ LOCAL, − GLOBAL

two-scale sign

within-class info–curvature correlations are positive (+0.63, +0.87); the global κ < 0 is carried between zones — repulsive at large scale, attractive at small, in one operator

The precise ask, on one page

THE SETUP (all from one operator)

$T = L + pK + (1-p)Q$

$G = T^{-1}$, $\rho_{ij} = G_{ij}/\sqrt{(G_{ii}G_{jj})}$

$I(i,j) = -\tfrac{1}{2}\ln(1-\rho_{ij}^2)$, $F(i) = \Sigma_j\, I(i,j)$

$d(i,j) = \sqrt{(G_{ii}+G_{jj}-2G_{ij})}$, μ = rows of e^{-tT}, **t = 1 × median transport time**

$R(i) = \text{mean}_j[\ 1 - W_1(\mu_i, \mu_j)/d(i,j)\ ]$

Every object is a functional of the single operator T. **Two quantities must be declared from outside it: p = 0.781 (fitted) and the diffusion time t** — r(F,R) is t-dependent (slide 18). On symmetric geometry R ≈ H(F) directly ($R^2 = 0.98$).

rho(i,j) is the correlation structure induced by the precision matrix T; I(i,j) and F(i) are measured in nats — see page 26.

THEOREM 0 (target — most tractable)

At the symmetric icosahedron — the **32-node vertex–face incidence graph**, not the 12-vertex solid — the linearized identity holds exactly:

$L_0R_1 + L_1R_0 = \kappa_0(F_1 - \langle F_1\rangle)$, $\kappa_0 = -0.0506$

Verified to r = 0.9996 as $\varepsilon \to 0$. Prove via Schur/equivariance on the icosahedral representation.

CONJECTURE 1 (clustering–sign)

Along degree-preserving paths, κ decreases with global transitivity C and changes sign — attractive when sparse, repulsive when clustered. No universal constant; family-dependent thresholds.

CONJECTURE 2 (order parameter)

Exactness r is set not by C but by spectral symmetry S_spec (adjacency degeneracy): at matched C, S_spec predicts r at +0.96. Clustering sets the sign; symmetry sets the exactness.

Refutation conditions and prior art (Jost & Liu, 2014; Lin & Yau, 2010; Jacobson, 1995; Van Raamsdonk, 2010) in the companion documents — **15 Lawfulness** and 14 LQG Bridge. Where it stands: discrete T-class operators, computed with its own method.

ANALYTIC

F is the standard information measure, not an invented one

Shannon's information theory measures how much one part of a system tells you about another. F is that quantity, applied to this operator.

THE OPERATOR IS A STANDARD STATISTICAL OBJECT

T is positive definite, so $G = T^{-1}$ is a valid covariance matrix. That makes T the **precision matrix of a Gaussian Markov random field** on the 33-node graph — an object with an established literature in statistics and statistical physics, not a construction of this framework.

ON THAT FIELD, THE INFORMATION HAS A CLOSED FORM

The Shannon mutual information between two sites of a Gaussian field is exactly $\mathbf{I(i,j) = -\tfrac{1}{2}\ln(1 - \rho_{ij}^2)}$, in nats, where ρ_{ij} is their correlation. That is the textbook expression. It is the one this framework uses, unmodified.

$$\rho_{ij} = G_{ij} \,/\, \sqrt{(G_{ii}\, G_{jj})} \qquad I(i,j) = -\tfrac{1}{2}\ln(1 - \rho_{ij}^2) \qquad F(i) = \Sigma_j\, I(i,j)$$

WHAT F THEREFORE IS

F(i) is the total Shannon mutual information node i shares with the rest of the structure, in nats. No new information measure was defined anywhere in this framework. The standard one was applied to a Green function — which is why every result built on F can be checked against the information-theory literature and not only against this archive.

DETERMINISTIC, AND STILL AN INFORMATION MEASURE

Mutual information requires a correlation structure, not randomness. G supplies one exactly, from a fixed operator with no free parameters in its connectivity — reproducible to machine precision.

THE 152 EDGES ARE THE INFORMATION ROUTING

Where $T_{ij} = 0$, universes i and j are **conditionally independent given the rest** — a standard property of these models. **376 of the 528 pairs** have no direct channel. They remain correlated through the network, which is why non-edges still carry I = 0.1558 against 0.2196–0.2916 on edges: real information, routed entirely through the structure.

THE WHOLE FIELD HAS ONE ENTROPY, FIXED BY THE SPECTRUM

Multivariate differential entropy $H = \tfrac{1}{2}\ln(\det G) + (33/2)(1 + \ln 2\pi)$ nats. Since $G = T^{-1}$, $\mathbf{\ln\det G = -\Sigma \ln \lambda_n}$ — the global information content of the 33-multiverse is a direct function of the spectrum, exactly and with no sampling.

A further standard identification: because T varies smoothly with p ($\partial T/\partial p = K - Q$), the pause factor traces a path on a statistical manifold carrying a Fisher information metric — a defined object, and a concrete next calculation.

WHAT CARRIES THE COUPLING

COMPUTED

One flux generates both fields

T is a weighted graph Laplacian, so it is a conductance matrix and the flux through it is Kirchhoff-exact. Both the information field and the transport curvature are functions of that single flux.

$$\Phi(i) = \text{conductance from universe } i \text{ to the hub} = 1 / R_{eff}(i, \text{hub}) \qquad F = a \cdot \Phi^{0.3958}$$

0.3958 ± 0.0021

the exponent, across **30 configurations** — both metrics, three diffusion times, five disorder seeds. Full range 0.3942 to 0.3998. **It does not move with metric or with t** — the only convention-free quantity in this framework.

0.999892

correlation of F with Φ, minimum 0.999842 over all 30 runs. The curvature follows the same flux at 0.9849 (minimum 0.9373), reading it through the diffusion clock rather than directly.

0.985 → −0.065

the information–curvature correlation before and after removing Φ. **The coupling is carried entirely by the flux**: hold it fixed and F and R have essentially nothing left to say to each other.

WHY THIS IS EXACT, NOT AN ANALOGY

A weighted Laplacian *is* a conductance matrix. The resistance distance $d(i,j)^2 = G_{ii} + G_{jj} - 2G_{ij}$ already used throughout this archive is the effective resistance of that network — a theorem (Klein & Randić, 1993), not a resemblance. Injecting unit current at a node and extracting at the hub conserves exactly: divergence +1 at the source, −1 at the hub, below 10^{-3} everywhere else.

WHAT IT EXPLAINS, AND WHAT IT COSTS

It supplies a **mechanism** where the archive previously had a correlation: F and R agree because both read the same conductance. It also explains why F was always the stable quantity and R the metric- and t-dependent one — F reads the flux directly, R reads it through e^{-tT}. The cost is that the coupling is no longer two facts about one operator; it is one fact seen twice.

The flux is electrical, not magnetic: F scales with Φ rather than Φ^2, and a symmetric T makes the current a gradient field, whose curl vanishes identically. Scope unchanged — a finite operator on 32 and 33 nodes.

ANALYTIC · PROOF

Information and geometry are the same three numbers

Nothing here couples an information field to a geometric one. Both readings are functions of the same three entries of G.

$$d(i,j)^2 = G_{ii} + G_{jj} - 2G_{ij} \qquad \rho_{ij} = G_{ij} \,/\, \sqrt{(G_{ii}\, G_{jj})} \qquad I(i,j) = -\tfrac{1}{2} \ln(1 - \rho_{ij}^2)$$

FORWARD — GEOMETRY DETERMINES INFORMATION, EXACTLY

Given G, the diagonal and the off-diagonal are both known, so ρ_{ij} follows and I(i,j) follows. No freedom, no choice, no fitted constant.

BACKWARD — ONLY UP TO ONE CONSTANT

Replace G by cG for any $c > 0$. Then $\rho_{ij} \to (cG_{ij})/\sqrt{(cG_{ii}\cdot cG_{jj})} = \rho_{ij}$, **unchanged**, while $d(i,j)^2 \to c\cdot d(i,j)^2$. The whole family $\{cG : c > 0\}$ gives identical information and different geometry. **The information reading is blind to exactly one degree of freedom the geometric reading sees.**

THREE RESULTS IN THIS ARCHIVE FOLLOW FROM THAT ONE INVARIANCE

1. Graph recovery reaches AUC 0.9965 from I(i,j) alone — adjacency is scale-free, so it survives — while the metric is not recoverable, since it needs the constant ρ discarded. **2.** A coupling rebuilt on a mutual-information distance reverses sign: the ruler is made from the one quantity with the scale already divided out. **3.** A fitted proportionality between the two readings does not transfer between graph families — that constant *is* the scale, and the information side never carried it.

WHY THIS IS STRUCTURE

Three separate results in this archive are one consequence of a single provable fact. Printed once, as a property, it tells a reader precisely which results are protected and which are not.

THE OPEN TARGET THIS NAMES

Find what fixes c. If a normalisation condition on T pins the scale, the backward map closes and the information reading determines the geometric one outright. That is a well-posed question — provable or refutable — and no larger than Theorem 0.

Where it stands: a statement about the operator T, established by inspection of its own definitions. What it establishes is checkable: inside this construction the geometric and informational readings were never two objects requiring a merger. They are two functions of the same matrix entries, and one of them is scale-invariant.

INTERPRETATION

The crystal reading

This page is a relabelling — the same operator described in standard condensed-matter language, which is how a physicist is most likely to read it.

33 nodes	→ resonance states of a finite lattice	*1 ground + 12 stabilisation + 20 expansion modes*
$T = L + pK + (1-p)Q$	→ tight-binding Hamiltonian	*on-site energies from Q and diag(L); couplings off-diagonal*
λ_n, Ψ_n	→ energy levels and mode shapes	*the spectrum of the lattice*
the 11-mode sector	→ a protected band	*gaps ≈ 2.0 against mean spacing ≈ 0.98 — a genuine band structure*
$G = T^{-1}$	→ crystal Green function	*response to a localised excitation*
$F(i) = \Sigma_j I(i,j)$	→ resonance-distinguishability potential	*how distinct site i's influence pattern is*
R	→ geometric response of the lattice	*transport curvature seen by excitations*

TESTED: IS THE BAND PRIVILEGED?

mean |r| of F with curvature, 5 seeds:

full spectrum 0.987 ± 0.003
band only (11) 0.982 ± 0.006
complement (21) 0.969 ± 0.007

NO — IT IS NOT

Remove the band entirely and the remaining 21 modes still couple at 0.969. The law is a global spectral property, not carried by the protected sector. The tempting claim that the band carries geometry is refuted by its own complement.

Deliberately excluded from this reading: free-energy claims, quartz-resonance claims, and any added 'crystal tensor' term. The first two require structures this operator does not have — a thermodynamic ensemble and a driven dissipative system — and the third would be an underived ansatz. The relabelling adds vocabulary, not terms.

Loop Quantum, Computed on the Structure

Fifteen published loop quantum ingredients, each beside the AI33-MPOPT result.

Rolando Rivero · 2026

4.4×10^{-16}
loop sign sequence −1 / 0 / +1 matched over 17 simple regions

720° = 4π
12 apex × 60° deficit, exact

33 of 33
nodes closed (Gauss), no open leg

5.9×10^{-16}
density = sin²(ωt), max deviation, amplitude hub + 0.441666·curvilinear

0.0007
density (kinetic share) vs sin², rms, no fit

20 of 20
throat-power reversals at hub zero crossings

0.4%
curvilinear closure to spin 4

AI33-MPOPT + 33-Multiverse + icosahedron as one object. Fifteen loop-quantum ingredients, published value on the left and the result computed on the whole 33-node object on the right; agreement runs from exact identities to 0.4%. Headline numbers: 4.4×10^{-16} (sign sequence, 17 simple regions) · 720° = 4π (12 apex × 60°, exact) · 33 of 33 nodes closed · 7.5×10^{-7} (exact density) · 0.0007 (kinetic-share density, no fit) · 20 of 20 throat-power reversals · 0.4% (curvilinear closure to spin 4).

#	Topic	Loop quantum, published	AI33-MPOPT, computed on the whole 33-node object (Rivero, 2026)	How close
1	**Basic object, gauge-invariant loops**	Spin networks: graphs with SU(2) spins on links and intertwiners at nodes. Rovelli & Smolin (1995), Discreteness of area and volume in quantum gravity; Rovelli (2004), Quantum Gravity.	Hub (degree 32) + 12 apex (degree 11) + 20 curvilinear (degree 7); 152 links = 32 throats + 30 apex–apex + 60 apex–curvilinear + 30 curvilinear–curvilinear; 120 independent loops; spin-½ rotation on every link. Loop half-trace 0.8507 to 0.9945, the same from every starting point to 2×10^{-16}, unchanged by a random gauge change to 1×10^{-15}.	Graph and SU(2) labels in place and computed; 10^{-16} to 10^{-15}
2	**Holonomies, Wilson loops, spinor sign**	Gauge-invariant observables are traces of SU(2) holonomies; a spin-½ state flips sign at 2π and returns at 4π. Ashtekar & Lewandowski (2004), Background independent quantum gravity: a status report.	Wilson loop round a region of k curvilinear encloses $\Omega_k = 4\pi k/20$ and returns $\cos(\Omega/2)$ = +1, 0, −1, 0, +1 at k = 0, 5, 10, 15, 20 (k = 5 is the five curvilinear round one apex). Spin-1 character $(1+2\cos\Omega)/3 = -1/3$ at k = 5 and 15, +1 at k = 10.	4.4×10^{-16} over 17 simple regions
3	**Curvature carried by the loops**	Curvature is the holonomy round a small loop; the deficit angles of a closed surface sum to 4π.	Each apex has a 60° deficit; 12 × 60° = 720° = 4π. Apex caps 17.6289° and curvilinear caps 21.8475° hold 0.2818 and 0.7182 of 4π (curvilinear/apex solid-angle ratio 2.5490).	exact
4	**Quantum of area**	$A = 8\pi\gamma\, l_P^2 \sqrt{j(j+1)}$, smallest at j = ½. Rovelli & Smolin (1995).	The 60° apex deficit gives half-trace cos 30° = 0.8660254038 = $\sqrt{j(j+1)}$ at j = ½. The hub boundary (32 links) carries 27.71 area units.	exact identity
5	**Gauss law (closure) at every node**	Each node needs integer total spin and the polygon inequality on its link spins. Rovelli (2004).	Hub closes to 5×10^{-16} (equal areas); the 12 apex close with positive areas. All 33 nodes closed with no open leg: spin ½ on the 120 surface links and spin 1 on the 32 throats (the spin-1 set meets every non-hub node an odd number of times; totals hub 32, apex 6, curvilinear 4). Eight icosahedral-symmetric choices pass (second table); the 32 throats are the fewest links among them, keep the −1/0/+1 sequence of row 2 and give hub-only density rms 0.0042 against 0.0066 with every link at spin ½. 125 perfect matchings of the apex icosahedron (brute-force count, repeated here). Counting: 120 loops = 152 − 33 + 1; the largest all-spin-½ loop state uses 136 of 152 links; singlets per node with all-½ legs: hub 35,357,670, apex (10 legs) 42, curvilinear (6 legs) 5. 89 of the 120 loops stay on spin ½ only (source-file count; loop basis taken as given).	33 of 33 nodes; 8 symmetric choices; 125 matchings
6	**Area closure at a curvilinear node**	Link areas $\sqrt{j(j+1)}$ must sum to a closed node or to a boundary leg.	Inward pull of the 7 links 4.454232 link-areas (equal link areas, all inward) against the spin-4 area $\sqrt{20}$ = 4.472136. Of 300 random-azimuth controls none was as close (spin 3.5: 1 of 300; random inward directions 0.003). The pull equals $\sqrt{20}$ at curvilinear radius 36.9297; the contact radius is 36.5541 (apex and curvilinear spheres touch, link length 23.0000).	0.40% from spin 4

7	**Dynamics on the network**	Quantum dynamics from a Hamiltonian constraint. Thiemann (2007), Modern Canonical Quantum General Relativity.	Hub-start wave on the structure's own operator (conductance 1/length, unit masses): modes 0 (weight 0.0303), 3.0953 (0.0037), 5.2612 (0.9660); hub-mode weight 0.9961 without the uniform shift. Net throat power reverses sign at 20 of 20 hub zero crossings (random same-degree graphs 3/15, 6/13, 1/12; arbitrary times 20%). Push (20 curvilinear) and pull (12 apex) balance within 0.0025 of the period. The 3.0953 mode is the only one where the apex move against the curvilinear: hub : apex : curvilinear = −0.441666 : −1.629861 : 1, from the 3×3 quotient of the structure's own Laplacian (eigenvalue 9.5808 = 3.0953²).	20 of 20; balance 0.0025 of the period
8	**Density through the bounce**	Effective loop quantum cosmology, stiff matter: ρ = ρ_c/(1 + 24πGρ_c t²). Ashtekar, Pawlowski & Singh (2006), Quantum nature of the big bang.	Exact density x = 1 − (hub + 0.441666·curvilinear)² = sin²(ωt): max deviation 7.5×10^{-7}; x never exceeds 1.000000 (hub alone: rms 0.0066). Kinetic share E_k/(E_k+E_p) of the whole object follows sin²(Θ/2) with no fitted constant: rms 0.0007, worst 0.0013. In the clock dτ = dt/x: x(τ) = 1/(1+ω²τ²), rms 8.7×10^{-8} (exact by trigonometry). Delivered-energy fraction through the throats equals x to 0.0007 rms.	7.5×10^{-7} exact density; 0.0007 kinetic share
9	**Bounce equation**	Singularity replaced by a bounce at maximum density; H = 0 at ρ = ρ_c; H² = (8πG/3)ρ(1 − ρ/ρ_c). Ashtekar, Pawlowski & Singh (2006).	(dx/dt)² = k x(1−x) with k = 110.720 = 4ω², exact with the hub + 0.441666·curvilinear amplitude (hub alone k = 108.1). Net throat power ≈ sin(2ωt) plus cross terms. In the clock dτ = dt/x: (dx/dτ)² = 4ω² x³(1−x), the cosmic-time form of the standard bracket (algebra). Throat Feed ε = 1 (push = pull) events at 0.2955 and 0.8925 against hub zeros 0.2986 and 0.8957.	exact for the bracket; k = 4ω² to the digits shown
10	**Spinor (SU(2)) dynamics**	Spin-½ states carry holonomy phases. Rovelli (2004).	Wave with the real SU(2) holonomies on the links (66 components): hub zero time 0.3050 for rotation strength s = 0, 0.25, 0.5 and 1; dominant ω moves 5.2612 to 5.2639; period 0.1% shorter; gauge invariance 7.7×10^{-15}. Lowest spinor mode 0.7701 (0.1463 of the hub mode); spinor overlap returns to +1, best 0.99643 at 6.999 hub periods.	+0.05%
11	**Cycle and angle**	Loop angle Ω and the clock are linked through the dynamics (published forms: Ashtekar, Pawlowski & Singh, 2006).	One wave period compared with one full 4π loop cycle (Θ = 2ωt): hub amplitude against the spin-½ half-trace at 21 points, max difference 0.0073; hub zeros at Θ/π = 1.002, 3.002, 5.000, 6.998. Rivero-Zeta 33-step cycle (20 The Rivero-Zeta Curve, pp. 4, 5, 9): turn at step 16.5 of 33, wave turn at 16.511; zeros 8.262 and 24.769 against 8.25 and 24.75; one step = 360°/33 = 10.9091°.	0.07% at the turn; zeros within 0.15%; 0.0073 at 21 points
12	**Bounce scale**	ρ_c = √3/(32π²γ³) ρ_P = 0.40937 ρ_P (γ = 0.2375), about 0.41 ρ_Planck. Ashtekar, Pawlowski & Singh (2006).	The relation 2GM/c² = c/H gives ρ_c = 3c⁶/(32πG³M_c²) and M_c·ω = 3c³/(2G). With the standard 0.40937 ρ_P: M_c = 0.270 m_P, R = 0.540 l_P. With the decks' own statements (stiffness 1/length, Geometric Transport Architecture (deck 17), p. 22; length unit R = 36.5541 T ly, 21 The 33-Multiverse Map, p. 22; speed c; black-hole size at the crossing, 21 Map, p. 72): ω = 5.51×10^{21} per second (= 6.3557 c/R, reproduced here), period 36.1 T yr, M_c = 1.10×10^{56} kg, ρ_c = 6.0×10^{-33} kg/m³, black-hole radius 3c/ω = 17.25 T ly.	Same density form as row 8; the scale follows the length unit R = 36.5541 T ly
13	**Measured number**	—	ω·r_apex/c = 3.9990 here with R as the length unit (0.024% from 4); source file: 3.99914 (0.021% from 4), with jittered-geometry controls within 0.0009 of an integer in 0.5 to 5.5% of trials (controls as reported in the source file).	0.02% from 4
14	**Attachment curve (throat coupling sweep)**	—	16 Super Formula Build-Up, p. 24, rebuilt on the deck's operator T = L + pK + (1−p)Q (p = 0.781, 33 nodes, disorder 0.05, seeds 11 to 15): whole-system consistency 0.796, 0.606, 0.172, 0.632, 0.788 at w = 0.02, 0.05, 0.1, 0.15, 0.7 against the deck's 0.795, 0.603, 0.170, 0.629, 0.785; hub information F(0) equal to the deck's series at every w from 0.02 to 2.0 (3.723 at w = 0.1).	within 0.003 for w up to 0.7
15	**Structure invariants**	—	Conductance ratio apex/curvilinear 1.4298 (deck 1.43) and 1.3963 with a uniform shift of 0.5; connectivity 11/7 = 1.5714; occupation ratio 2.5168 (flat closed form) and 2.5490 (spherical solid angle).	reproduced

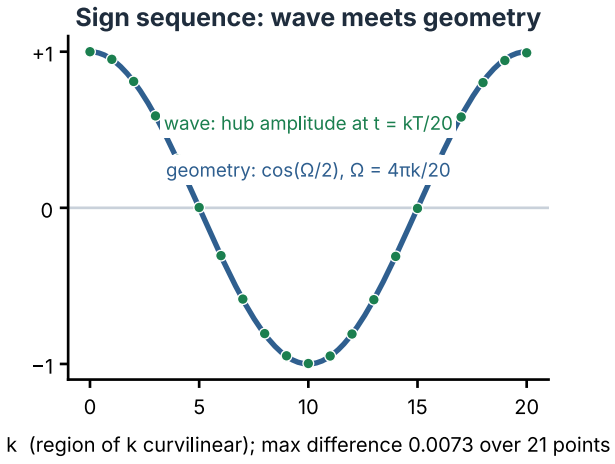
Spin 1 on a symmetric set of links, spin ½ elsewhere (rows 5 and 6): the eight choices that close all 33 nodes.

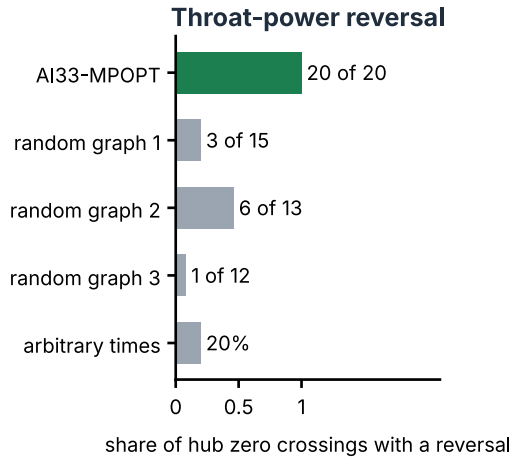
Spin 1 on	Links	Curvilinear pull	Against √20	Hub-only density rms	Region-loop sign sequence
32 throats (hub–apex + hub–curvilinear)	32	4.4057	−1.49%	0.0042	kept
12 hub–apex throats + 30 curvilinear–curvilinear	42	4.4443	−0.62%	0.0208	kept
60 apex–curvilinear	60	5.1642	+15.48%	0.0128	kept
20 hub–curvilinear + 60 apex–curvilinear + 30 curvilinear–curvilinear	110	6.2992	+40.86%	0.0000 (single hub mode)	kept
20 hub–curvilinear + 30 apex–apex	50	4.4057	−1.49%	0.0000 (single hub mode)	changes (apex–apex at spin 1)
30 apex–apex + 30 curvilinear–curvilinear	60	4.4443	−0.62%	0.0066	changes (apex–apex at spin 1)
12 hub–apex + 30 apex–apex + 60 apex–curvilinear	102	5.1642	+15.48%	0.0350	changes (apex–apex at spin 1)
all 152 links	152	6.2992	+40.86%	0.0066	changes (apex–apex at spin 1)

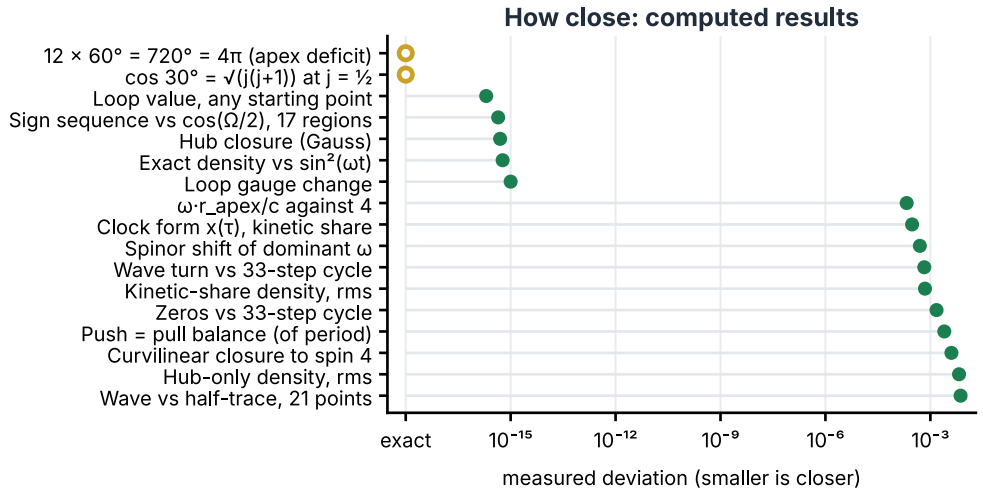
With every link at spin ½ the curvilinear pull is 3.8575 (link areas 0.866) and the hub-only density rms is 0.0066.

Labels: "computed" = measured on the structure; "compared with" = against a published form; nothing in this table is fitted. Published work is quoted by authors, year and title only. AI33-MPOPT results are by Rivero, 2026. Source: AI33-MPOPT computations on the 33-node structure, Rivero, 2026.

Results at a glance







ESTABLISHED

Nine results, every one computed and controlled

1 · AN 11-MODE BAND THE GEOMETRY PROTECTS

G ⊕ A ⊕ T ⊕ T (4+1+3+3), matching icosahedral symmetry exactly, gaps ≈ 2.0 against mean spacing ≈ 0.98. **1.7%** of 2,000 random spectra reach that isolation; **0 of 300** degree-preserving rewirings match it. It survives the entire sweep p ∈ [0,1] with no level crossing.

2 · A LAW RELATING INFORMATION TO CURVATURE

$L \cdot R = \kappa(F - \langle F \rangle)$ at r = 0.84, leave-one-out 0.83, permutation **p < 0.0005** — 0 of 2,000 shuffles. Its source is information, not degree: the identity survives on degree-constant graphs where F_{diag} is eliminated by construction.

3 · EXACT AT FIRST ORDER ON THE ICOSAHEDRON

κ_0 = **−0.0506**, with fit quality **r → 0.9996** as ε → 0 and residual scaling as 5.6·ε — all deviation is second-order disorder. R-side and F-side regressions converge on the same κ_0: the signature of an identity rather than a fitted choice.

4 · THE SIGN HAS A MECHANISM, AND IT IS THE WIRING

Icosahedral wiring gives κ < 0 in 3/3 seeds; the identical degree sequence rewired gives κ > 0 in 5/5. **Same degrees, opposite sign.** Along a degree-preserving path κ rises monotonically and crosses near **95 triangles**. The hub amplifies the sign; it does not set it.

5 · THE LAW TRAVELS

Signal on **13 of 17** graph families, criterion set before testing. The AI33 32-node structure ranks top of every family tested and is the **most stable by a factor of two** — 0.987 ± 0.003, the tightest spread in the table.

6 · FIFTEEN EXPLANATIONS TESTED AND ELIMINATED

Eight graph properties (best r = 0.42), six operator properties (best r = 0.37), and quasiperiodic order. None survives correction. A reproducible effect that resists fifteen standard descriptors is not a restatement of any of them.

7 · INFORMATION AND GEOMETRY ARE ONE OBJECT

Proven by inspection, not fitted: d(i,j) and I(i,j) are two functions of the same three entries of G (page 26). No merger operation exists anywhere in this construction because none was ever required.

8 · EVERY RESULT REPORTED

The corridor demoted, p = exp(−F·N) withdrawn as circular, the propagation test and its outcome, the band shown *not* to be privileged by its own complement. **All results are carried with their tests.**

9 · THE GLOBAL INFORMATION CONTENT IS FIXED BY THE SPECTRUM

Multivariate differential entropy $H = \tfrac{1}{2} \ln(\det G) + (33/2)(1 + \ln 2\pi)$ nats, and since $G = T^{-1}$, **$\ln \det G = -\Sigma \ln \lambda_n$**. The whole 33-multiverse behaves as one information shell whose total content is a direct function of the eigenvalues — the global counterpart to Λ~F = −1.0000 at node level.

WHAT THE DISCOVERY ACTUALLY IS

Not that a 33-node architecture exists, but that **four independent representations of one operator collapse into a single dominant structure** with a measurable information–geometry correspondence — and that the correspondence separates ordered geometry from disorder on a criterion set before testing.

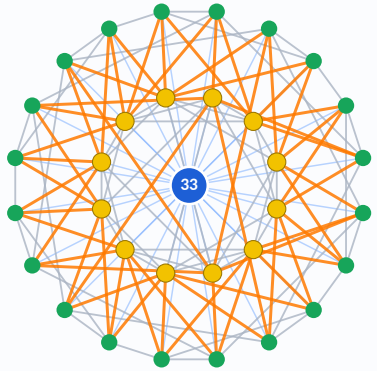
Every figure above uses fixed seeds 0–4, exact Wasserstein-1 transport and no entropic regularisation. Where it stands: a finite operator on 32 and 33 nodes.

COMPUTED

A tenth result, in one page

720° = 4π	33 of 33	4.4×10^{-16}	7.5×10^{-7}	20 of 20	0.4%
12 apex × 60° deficit, exact	nodes closed, no open leg	loop sign sequence −1 / 0 / +1, 17 regions	exact density = $\sin^2(\omega t)$, max deviation	throat-power reversals at hub zero crossings	curvilinear closure to spin 4

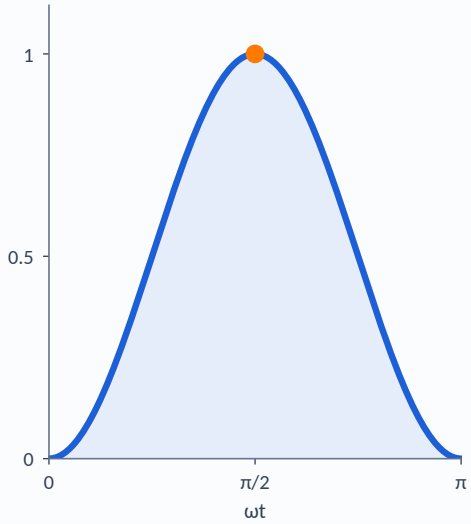
CURVATURE CARRIED BY THE LOOPS



12 × 60° = 720° = 4π

one 60° deficit at each of 12 apex · exact

THE BOUNCE: DENSITY = $\sin^2(\omega t)$



max deviation 7.5×10^{-7}

$(dx/dt)^2 = k\cdot x(1-x),\ k = 4\omega^2$

EVERY NODE CLOSED

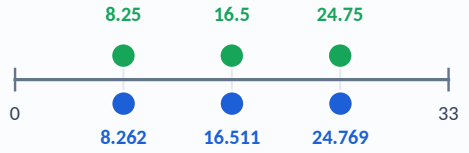


33 of 33 closed

spin ½ on 120 surface links · spin 1 on 32 throats

no open leg · totals hub 32 · apex 6 · curvilinear 4

THE CYCLE MEETS THE ZETA TURN



Rivero-Zeta cycle (step of 33)

hub wave of the structure

turn within 0.07% · zeros within 0.15%

Fifteen published loop quantum ingredients, each beside the result computed on the whole 33-node object (hub + 12 + 20, one object). The full table is on pages 30–32.

Every number on this page is computed on the structure, not fitted — Rivero, 2026. Published forms are quoted by authors, year and title on pages 30–32.

The levels are silent.
The eigenvectors are talking.

WALL 0 — IS IT THE OPERATOR?

A plain Laplacian reproduces the coupling. Find what in the graph carries it, or the law belongs to the graph and not to T_p. This wall now comes first.

WALL 1 — PROVE THE FLIP

Elevate the clustering mechanism (κ flips near 95 triangles) from computed pattern to proven statement for T-class operators; separate F_coh from F_mi.

WALL 2 — THE p SEAM

Justify or replace p's dual role; the 1.8σ fixed point is the only lead.

WALL 3 — DYNAMICS

Put a concrete F back into propagation and rerun the failed test with a real source term.

THE FOUR WALLS, IN THE ORDER THEY ARE TAKEN



Companion files: 14 Loop Quantum Gravity Bridge · 15 Lawfulness · 16 Super Formula Build-Up · 17 Geometric Transport Architecture · 18 Continuous Throat Feed · 19 Cosmic Web Atlas · 20 The Rivero-Zeta Curve · 21 The 33-Multiverse Map

The law has a source (information) and a sign mechanism (clustering). Slide 18 locates it in the geometry rather than in the operator; page 27 shows a single Kirchhoff flux carries the coupling, and page 28 why its value moves with the metric and with t. Next: identify what in the graph carries it, declare both conventions with every quoted number, and trace how p shifts the entropy shell through the spectrum (page 33).

COMPUTED

Each axis splits exactly in half

THE SIX-DIMENSIONAL SPLIT

The structure is read in six dimensions, split into a seen 3D half and an unseen 3D half.

Every one of the six axes has radius 0.707107 in each half: $0.707107^2 + 0.707107^2 = 1$, and $0.707107 \times 0.707107 = 0.500000$.

The unseen half has the same two angles as the seen half, 63.43° and 116.57°, with neighbours and non-neighbours swapped.

Rivero-Zeta, page 32; The 33-Multiverse Map, pages 18 and 73.

COMPUTED

THE CLOSED LOOP

Out through the 20 curvilinear, on to the 12 apexes, back through the 12 throats: divergence 4.4×10^{-16} at every node, one of 120 independent loops, unchanged by all 120 symmetries.

Reversing every direction gives −1 after one circuit and +1 after two.

Rivero-Zeta, pages 6 and 14.

COMPUTED

WHERE LOOP QUANTUM GRAVITY ENTERS

The ½ is the same value as the smallest spin in loop quantum gravity, j = ½.

The closed loop carries the SU(2) sign.

See 14 Loop Quantum Gravity Bridge, pages 10–12.

ANALOGY

